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Molecular insights into the nanoconfinement effect on the structure and dynamics of ionic liquids in carbon nanotubes†

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The properties and applications of ionic liquids (ILs) have been widely investigated when they are confined within nanochannels such as carbon nanotubes (CNTs). The confined ILs exhibit very different properties from their bulk state due to a nanoconfinement effect, which plays an important role in the performances of devices with ILs. In this work, we studied the effect of the charge carried by CNTs on confined ILs inside CNTs using molecular dynamics simulations. In charged CNTs, cations and anions are distributed separately along the radial directions, and the transition of orientations of the cations between parallel and vertical to CNTs occurs by changing the charge state of CNTs. The number of hydrogen bonds (HBs) formed by the confined ILs can be reduced by switching the surface charge of CNTs from positive to negative due to the contact modes between cations and anions as well as the distributions of cations in CNTs. The diffusivities along and vertical to the axial direction of CNTs were found to be non-monotonic owing to the "trade-off" effect from both ion pair interlocking and anchoring ILs on the CNT walls. Additionally, the region-dependent dynamics of ILs were also related to the intermolecular interactions in different regions of CNTs. Furthermore, the vibrational modes of ILs were obviously influenced in highly charged CNTs as determined by calculating the density of vibrational states, which demonstrated the transitions in the structure and interactions. The density distributions changed from single layer to double layers when increasing the pore size of neutral CNTs while the hydrogen bonds exhibited a non-monotonic tendency *versus* the pore sizes. Our results might help to understand the structure and dynamics of confined ILs as well as aid optimizing the performance of devices with ILs.

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1. Introduction

Ionic liquids (ILs) are a type of liquid materials composed of pairwise ions, including asymmetric organic cations and organic/inorganic anions, with melting points below 373 K. They are formed by pairwise ions where one is a large organic cation and the other is an organic/inorganic anion. ILs can exhibit several crucial physicochemical properties such as an extremely low vapor pressure, good thermostability, quite wide electrochemical window, and reasonable ion conductivity, and the properties can be tuned by coupling different species of cations and anions.^{1,2} The diverse natures of ILs open up a vast array of potential applications across various fields, including

energy storage,^{3,4} CO₂ capture and conversion,⁵ flexible sensors,⁶ and propellants.⁷ In various applications such as energy devices, ILs are typically assembled with carrier materials and transformed into solid devices to guarantee sustainable and stable operation.² To maintain the accessibility of ILs within solid devices simultaneously, porous materials such as porous carbon,^{8–10} metal-organic frameworks (MOFs),¹¹ and silica¹² are often employed. Porous carbon materials such as graphene and carbon nanotubes (CNTs) exhibit remarkable conductivity and customizable pore sizes and structures, rendering them as promising application candidates, particularly in energy storage.

Owing to the presence of nanoconfinement effects, there are significant differences in the properties between confined and bulk ILs; for instance, abnormal phase transition and wetting behavior, layer structure near the interface, variation of mobility, and so on.² Moreover, the electrostatic interaction between the surface and ILs also has a further impact on the behavior of ILs if the nanochannels are charged in some circumstances.¹³ The performance of IL devices often depends on the state of the ILs, so understanding the microscopic characteristics and

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mechanisms of ILs confined in charged nanochannel at the molecular level is of great importance. To unravel the mysteries of ILs within nanochannels, both advanced experimental characterization techniques and theoretical simulation methods have been adopted in previous research. Some experimental techniques to investigate the confined ILs include a range of *in situ* and *ex situ* methods, like nuclear magnetic resonance (NMR),^{14,15} X-ray scattering,^{16,17} electrochemical quartz crystal microbalance (EQCM),^{18,19} and atomic force microscopy (AFM).²⁰ For instance, EQCM was adopted to qualitatively characterize the behavior of the IL 1-ethyl-3-methyl imidazolium bis(trifluoromethanesulfonyl)imide ([EMIM][TFSI]), in which single-layer graphene (SLG) was immersed, and the charge on the SLG electrode.¹⁸ It was demonstrated that cations would be repulsed by the interface region with the potential varying during electrode scanning from the potential of zero charge (PZC) to a positive state. Conversely, while scanning from the PZC to a negative state, the orientation of anions close to the interface changed, which caused the conversion of the ionic packing mode. Zhou *et al.*²⁰ used electrochemical three-dimensional (3D) atomic force microscopy (EC-3D-AFM), developed in-house, to explore the structure of ILs confined between graphene, and observed discrete layers of ions in the electric double layer (EDL) with quasi-periodic, zig-zag-like oscillations. Furthermore, research on the structure of ILs confined between slit-like carbide-derived carbon (CDC) nanochannels was performed with X-ray scattering, which revealed a partial breakage of the Coulombic-ordering structure.¹⁷

In addition to the aforementioned experiments, theoretical simulation approaches, like molecular dynamics (MD) simulations and Monte Carlo (MC) simulations, have been carried out to understand the mechanism of ILs inside nanochannels. Many properties of confined ILs can be predicted by simulations, such as the lamellar structure and regular orientation of ions adjacent to the interface, and these have been found to be in good accordance with experimental results. Following the text above, Futamura and co-workers¹⁷ used the MC method to support their discovery from experiments with X-ray scattering, and the similar results obtained *via* these two methods proved that the super-ionic state indeed existed. Wang *et al.* reported that the structural and thermodynamic properties of confined ILs varied in a non-monotonic way with the variation of the pore size, as elucidated by combined analyses of the hydrogen bond (HB) networks within ILs and the external pressure from nanochannel, as well as electrostatic pressure between ions.²¹ Kondrat and co-workers revealed that the dynamics of ILs varied non-monotonically with the charge density of electrodes when the pore size was fixed.²² Likewise, Mo *et al.* found a non-monotonic relationship between the pore size and diffusion of ILs when the charge density of electrodes was fixed.²³ Both research studies above demonstrate that the dynamics of ILs confined inside nanochannels are impacted by multiple factors, and also suggest that the dynamic properties of confined ILs strongly depend on structural properties. While most studies about the dynamics of confined ILs are based on the confinement in 2D slit pores, it would be meaningful to carry

out relevant research in non-planar nanochannels, like carbon nanotubes (CNTs).

In the present work, extensive MD simulations were performed to explore the structural and dynamical behaviors of ILs under nanoconfinement by a single-walled CNT. We focused on how the charge densities of CNTs influence the properties of ILs, as well as the relation between the structure and dynamics of confined ILs. We chose a widely used imidazolium-based ionic liquid, namely 1-butyl-3-methyl imidazolium tetrafluoroborate ([BMIM][BF₄]), and armchair CNT nanochannels for study. Charges were distributed on the CNT uniformly before the simulation, so-called the constant charge method (CCM). The structural and dynamical properties of ILs confined in CNTs with different pore sizes and surface charge densities were then investigated.

2. Methods

In order to explore the properties of confined ILs, a simulation system containing both a CNT nanochannel and ILs was constructed, as shown in Fig. 1. The nanochannel included two parts: armchair-type single-walled CNT and monolayer graphene sheets attached to the ends of both sides of CNT to prevent ILs from approaching the CNT from outside. The chosen ILs [BMIM][BF₄] were placed on both sides of the graphene sheets, being allowed to enter the CNT after the simulation started. We distributed 600 IL pairs in the system with a neutral CNT, and adjusted the number of cations or anions in the systems with positively or negatively charged CNTs, respectively, in order to guarantee the whole systems were electroneutral. Besides, the positions of all the carbon atoms in the electrode systems were fixed during the simulation. We mainly studied ILs in the CNTs with a chiral index of

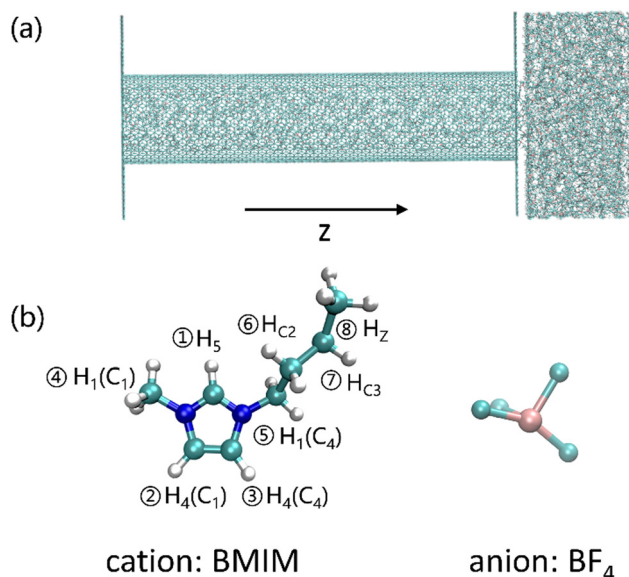


Fig. 1 (a) Front view of the simulation system, and (b) structures of the cation BMIM and anion BF₄; where the number and the labels around the cation represent the H atoms in different groups in the cation.

20, which means that the radius of the channel was about 1.337 nm. In the study on the pore size effect, seven types of CNTs with different sizes were adopted, namely (7,7), (9,9), (10,10), (11,11), (12,12), (13,13), and (15,15) CNTs. The length of CNTs was kept at 11.984 nm. The ILs located in the middle 10 nm region of CNT were collected for data analysis, in order to avoid the influence of the edge between the CNT and graphene sheet.

All the intermolecular and intramolecular interactions considered in the system, such as bond, angle, dihedral, van der Waals and Coulomb interactions, and so on, were modeled by the force field based on the AMBER framework as adopted in previous works.^{24–27} The nonbonded interactions between the particles were described through dispersion terms and electrostatic terms, respectively. The Leonard–Jones (LJ) potential was adopted to compute the dispersion interactions, while the long-range Coulomb interactions were computed *via* the particle-mesh Ewald (PME) algorithm.²⁸ The LJ potential was truncated by setting the cutoff distance of 1.6 nm in order to improve the simulation efficiency. The periodic boundary condition (PBC) was carried out in all three dimensions to diminish, as far as possible, the boundary effect caused by the finite size of the simulation system.

The charges were distributed uniformly on each carbon atom of the CNT to simulate the charged electrode, by the so-called CCM method.^{20,21,29} In addition to neutral CNT, four groups of CNTs with different charge densities, including $\pm 0.5 \text{ e nm}^{-2}$ and $\pm 1 \text{ e nm}^{-2}$, were set up to study the relationship between the IL behavior and the surface charge density. In comparison with the CCM, the constant potential method (CPM), which takes account of the polarization of the electrode, is regarded as a more realistic way to mimic the electrode currently.^{30,31} However, the calculations of charge distribution and total charge of the electrode to maintain the potential are very time consuming, resulting in a higher calculative cost. Prior studies have indicated that a major shortcoming of the CCM is primarily evident in assessing charge dynamics during the capacitor charging process,³² whereas the assessment of confined ILs in an equilibrium state through qualitative analysis has yielded outcomes that align with those obtained *via* the CPM approach.^{33,34} As a consequence, the CCM method is still feasible and offers a good balance of simplicity, speed of simulation, and accuracy.

All the MD simulations in this work were run with the GROMACS software package.³⁵ To generate the simulation system, bulk ILs with random configurations were equilibrated for 10 ns and then packed with the electrode *via* the program PACKMOL.³⁶ Subsequently, the simulations were carried out for 10 ns in the NPT ensemble to equilibrate the system, in which the temperature was set to 400 K and maintained *via* a Berendsen thermostat and the pressure was kept at 1 bar through Berendsen pressure coupling applied in only the axial direction of the CNT. After that, the simulations were further conducted for 40 ns to generate trajectories in the NVT ensemble, while the temperature was controlled at 400 K by a Nosé–Hoover thermostat. In such a way, the properties of the ILs could be calculated from the trajectories

of the 40 ns production simulations. All the simulations were run at least 5 times separately in order to get parallel data for averaging the calculated properties.

3. Results and discussions

3.1 Structure of ILs in CNTs with different charge densities

The first section of this work focuses on the structure and dynamics properties of ILs under confinement in cylindrical CNTs with different charge densities. The chiral index of the CNT was maintained as (20,20). First, the structural properties of the confined ILs in CNTs were characterized by calculating the number density along the radial direction. Fig. 2(a)–(e) show the density profiles of the centers of mass (COM) of both the cation BMIM and anion BF_4 in the CNTs with different charge densities, wherein the imidazole rings and butyl tails of the cation were considered separately by calculating the COMs of the heavy atoms (C and N) in the rings and tails, respectively. The mass density profiles of the ILs in the CNTs with different charge densities are available in Fig. S1 in the ESI,[†] together with a comparison with the bulk densities of the ILs. The density oscillations in Fig. 2 demonstrate that the ILs formed a layer structure. The layer structures of the ILs exhibited heterogeneously in the charged CNTs; that is, as the charge densities increased to $\pm 0.5 \text{ e nm}^{-2}$, the counter-ions (ions with a counter charge to CNT) were distributed more clearly in the regions near the tube walls, accompanied with the growth of a density peak next to the tube wall. In contrast, the density peaks of the co-ions (ions with the same charge as CNT) were shifted toward the central axes of the CNTs at $\pm 0.5 \text{ e nm}^{-2}$, forming peaks that alternated with those of the counter-ions subsequently. When the charge densities were further increased to $\pm 1 \text{ e nm}^{-2}$, the co-ions from different layers accumulated and merged into a single layer clamped by the double layers of counter-ions from both sides. The density oscillation of the confined ILs was induced by the joint effect of several sorts of interactions, among which the Coulombic interactions between ions as well as between ions and CNTs had superior intensity and took the leading roles. Specifically, the dense ions gathered within the first density maximum closest to the tube walls led to the emergence of potential energy peaks in the neutral CNT. Such an energy landscape would subsequently act on other nearby ions, inducing the formation of a second density maximum at a certain distance to the first one, as well as a density valley between the two peaks. But if the CNTs carried charges, the net electrostatic interactions between the CNTs with counter-ions and the co-ions would be attractive and repulsive, respectively, resulting in a heterogeneous layer structure. This phenomenon concurred with the experimental research³⁷ and simulations performed *via* the constant potential method.³⁴

The local environment of ILs depends on their different locations inside the CNT. For instance, the IL layers close to the tube wall interact with other ILs from just one side; in other words, from the central axis direction of the CNT, while the other side is exposed to the tube wall; on the other hand, the

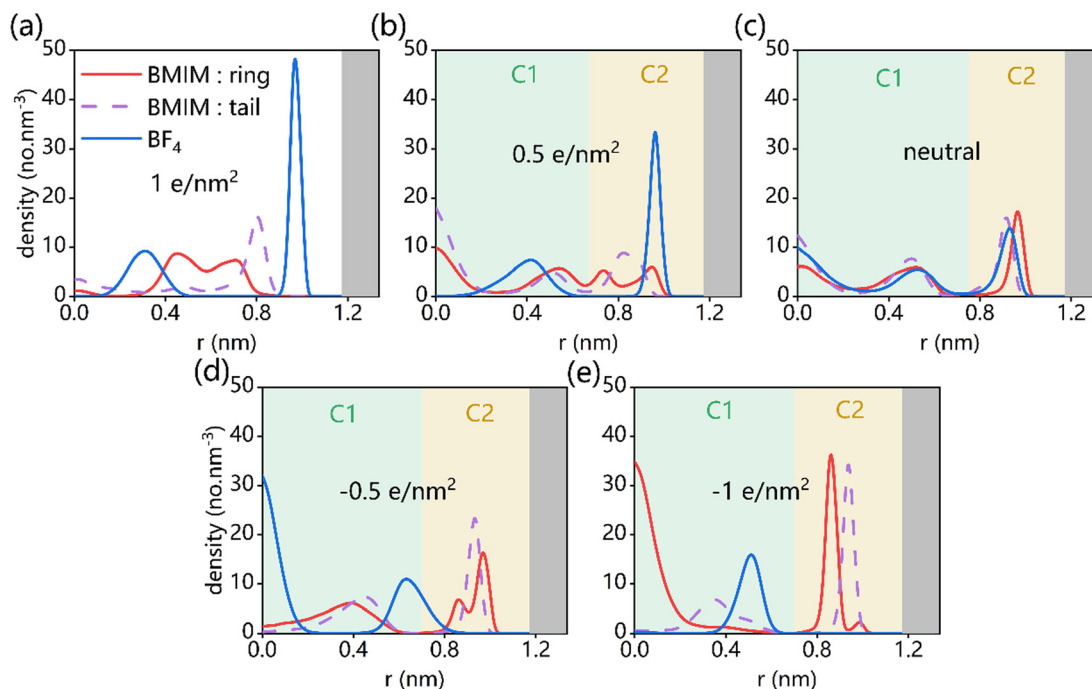


Fig. 2 Number density profiles of the IL [BMIM][BF₄] in the five CNTs with charge densities of: (a) 1, (b) 0.5, (c) 0 (neutral), (d) −0.5, and (e) −1 e nm^{−2}, respectively. Different background colors represent different regions based on the distribution of cations, where the green color means the central region (Region C1), and the yellow color corresponds to the boundary region (Region C2).

ions in the central regions are surrounded by other ions. In addition, the construction of different IL layers is distinct as well; the ions near the tube wall construct ring structures coaxial with the CNT; however, the ions away from tube walls are arranged more homogeneously. Thus, it is inferred that the ions belonging to diverse layers perhaps vary in both structural and dynamic properties owing to the discrepancy in their local environment and construction, so it is necessary to take apart the layers and study them separately. According to the arrangements and locations of layers formed by the confined cations and anions, the CNTs were divided into two groups of regions, with one group based on the density distribution of cations and the other one based on anions. In the former, the region next to the tube walls was labeled as Region C2 (yellow shaded area in Fig. 2(a)–(e)), while the other region containing the rest of the ions out of Region C2 was delimited as Region C1 (green shaded area in Fig. 2(a)–(e)), and the two regions were demarcated by the density valley next to the first maximum to the tube walls. Likewise, the group of regions of anions was delimited in a similar way. In the following text, Region C1 and Region A1 are to be recognized as central regions, while Region C2 and Region A2 are boundary regions. The demarcations of the two groups were distinct in CNTs with different charges, so the divided regions of cations are shown in Fig. 2(a)–(e), and those for the anions are shown in Fig. S2(a)–(e) (ESI†).

To further explore the orientation of ions in the nanotube, the orientation order parameter of cation $P_2(\cos \theta)$ was examined by the Legendre polynomial as follows,

$$P_2(\cos \theta) = 1.5(\cos \theta)^2 - 0.5 \quad (1)$$

where two groups of vectors were defined to describe the orientation, which are $\mathbf{r}_1$ and $\mathbf{r}_2$, $\mathbf{r}_3$, and $\mathbf{r}_4$ (Fig. 3(a) and (b)), respectively. The angle θ_1 between the vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ describes the orientation of the imidazole rings, and the angle θ_2 between $\mathbf{r}_3$ and $\mathbf{r}_4$ is calculated for the orientation of butyl chains. The value of $P_2(\cos \theta_1)$ close to 1 means that the imidazole rings lie parallel to the wall of the CNT, while the value of −0.5 means that the rings are perpendicular to the wall. For $P_2(\cos \theta_2)$, the value of 1 indicates that the butyl tails are vertical to the wall, while the value of −0.5 means a parallel orientation. The normalized orientation probability distribution functions (PDF) of the imidazole rings and butyl tails are shown in Fig. 3(c) and (d). Most imidazole rings tend to orient parallel to the tube wall in the neutral CNT. But if the charge densities of CNTs are increased to ± 0.5 e nm^{−2}, the proportion of imidazole rings with parallel orientation is reduced, and a considerable proportion of them turn to be perpendicular to the wall; besides, there were also some rings with a random orientation. These phenomena indicate that the orientation of the imidazole rings are more disordered in such surface charge densities than in neutral CNTs. When the charge densities on the CNTs were further increased to ± 1 e nm^{−2}, most imidazole rings turned to perpendicular to the tube wall, demonstrating that the highly ordered structure of imidazole rings with a perpendicular orientation replaced the disorder again as the absolute surface charge density increased (Fig. 3(c)). In contrast, the influence of the surface charge of the CNTs on the orientation of the butyl tails was inapparent, as only a small proportion of butyl tails tended to point out from the tube wall in the positively charged CNTs (Fig. 3(d)). According to our

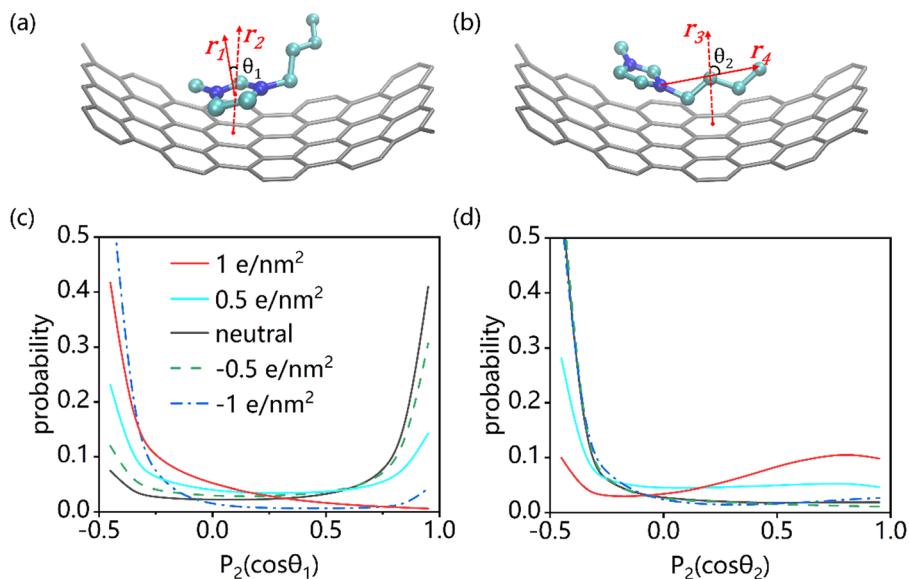


Fig. 3 θ_1 , θ_2 represent the angles between the unit vectors perpendicular to the tube wall and the vectors defining the orientation of imidazole rings and butyl tails, respectively (a) and (b). Normalized orientation probability functions (PDF) of the orientations of the (c) imidazole rings and (d) butyl tails.

previous study, the partial charge and electrostatic potential of each group in cations are distributed heterogeneously; for instance, the H₅ atom is more polar than the other two H atoms bonded to the imidazole ring.²⁶ Thus, the inhomogeneous interactions between the charged CNT and each group of cations result in a transition of the orientation of the imidazole rings. In addition, the interactions between the charged groups in the cations and anions from different layers in the CNTs could also contribute to the change in the orientation.³⁸ Fig. 4(a)–(d) show the normalized orientation of the PDFs of the imidazole rings in diverse regions defined in the preceding text and Fig. 2. In CNTs with electroneutrality, the imidazole rings in the boundary regions (C2) were oriented parallel to the wall significantly, while they were more disordered in the central regions (C1), which was attributed to the diverse interactions between the CNT and ions in different regions. In CNTs with $\pm 0.5 \text{ e nm}^{-2}$, the proportions of imidazole rings in the boundary region parallel to the tube walls were smaller than those in the neutral CNT, demonstrating that the orientation of imidazole rings located in this region presented some diversity; but almost all the imidazole rings were oriented perpendicularly after the absolute charge density increased to 1 e nm^{-2} (Fig. 4(d)), and only the negatively charged CNT (-1 e nm^{-2}) was considered, as there was no clear group of regions based on cations in the CNT with $+1 \text{ e nm}^{-2}$, in both the boundary and central regions. The orientation profiles of the imidazole rings as a function of the distance from the central axes of the CNTs are plotted in Fig. S3 (ESI[†]), which also illustrates that both the distribution of ions and charge density of the CNTs affect the orientation of cations. The changes and distributions of the orientations of imidazole rings were similar to the results obtained for ILs on graphene by Lynden-Bell and co-workers, which might also have an effect of screening the surface charge of the CNT wall, besides the electric double-layer structure with counter-ions and co-ions.^{39,40}

In addition, the division of the density of peaks of the imidazole rings and butyl tails in Fig. 2 at different surface charge densities also depended on the orientation of the imidazole rings. The COMs of the imidazole rings were almost in the same plane as the other atoms on the rings in the neutral CNT, since the imidazole rings kept almost planar throughout the whole simulation process; while the COMs of the butyl tails were non-coplanar with the imidazole rings, leading to a splitting of the density peaks of the tails and rings. In addition, the differences in polarities between the imidazole rings and butyl tails also resulted in the differences in their density peak positions. For instance, in Region C2 of the CNT with -1 e nm^{-2} , the COM of the rings was located further away from the wall than for the butyl tails (Fig. 2(e)), as the rings were perpendicular to the tube wall while the tails lay parallel to the wall, resulting in the density peak of the tails being closer to the wall; while in the CNT with $+1 \text{ e nm}^{-2}$, the division of tails and rings was more significant, with the latter further away from the wall, which could probably be attributed to the stronger electrostatic repulsion of the imidazole rings with the positively charged tube wall, as well as the orientation conversion of some tails from parallel to oblique. For the CNTs with lower charge densities of $\pm 0.5 \text{ e nm}^{-2}$, the density distribution of rings in Region C2 was split into two peaks around the tail due to the more random orientation of the rings.

The charge on the CNT led to the rearrangements of ion layers and the conversion of the orientation of the cations, which were likely related to the way the ion pair contacted and interacted with each other. The radial distribution functions (RDFs) between the COMs of the cation BMIM and anion BF₄ are plotted in Fig. 5(a). The positions of the peaks in the RDFs reflect the spatial relationship between the cations and anions and show it does not strongly depend on the charge on the CNTs, except for the CNT with -1 e nm^{-2} , where the position of

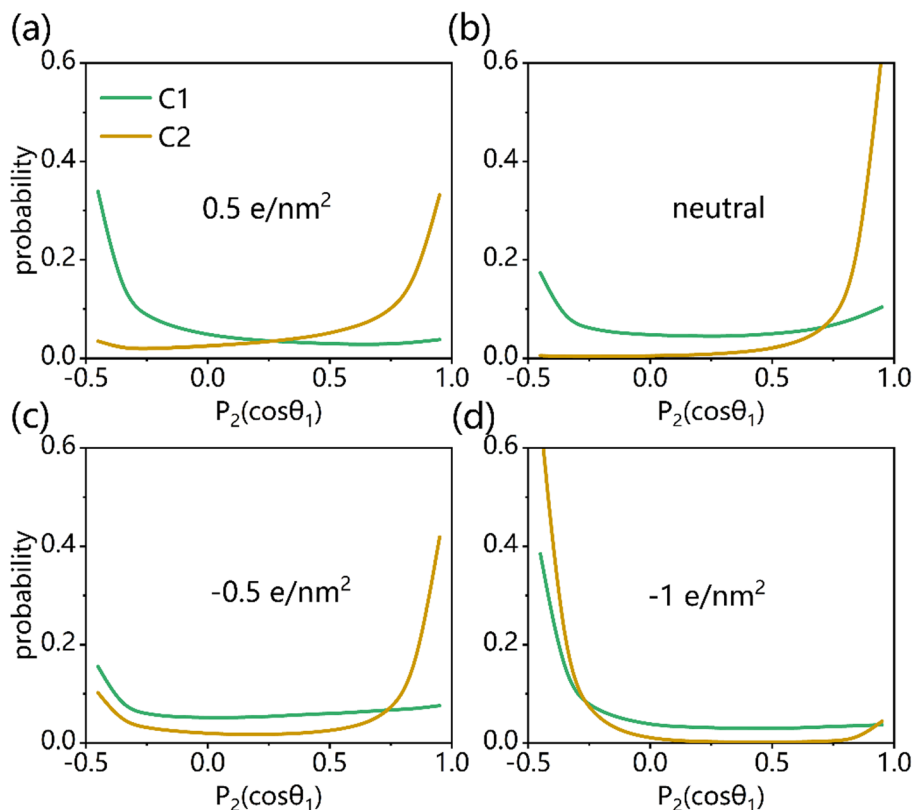


Fig. 4 Normalized orientation PDFs of the imidazole rings of the cations in different regions in the four CNTs with charge densities of (a) 0.5, (b) 0 (neutral), (c) -0.5 , and (d) -1 e nm^{-2} , while the cations in the 1 e nm^{-2} CNT could be ignored due to there being no obvious division of cations in the different regions.

the first peak in the RDF curve was shifted slightly to the right, which may be related the large difference between the number of cations and anions (Fig. 5(e)), as well as the larger volume occupied by the cations. In contrast, the RDFs between ions with like charges (BMIM–BMIM, BF_4 – BF_4) obviously changed with the charge densities of the CNTs (Fig. 5(b) and (c)); whereby the position, shape, and the numbers of peaks in the RDFs of both the cation pairs BMIM–BMIM and anion pairs BF_4 – BF_4 were altered as the charge densities were modified, reflecting the complicated changes in the microstructure of the ILs. In addition, the first and second peaks of the RDF between the cations and anions were located at 0.5 nm and 1.2 nm in the neutral CNT, respectively, while both the nearest peaks of RDFs based on BMIM–BMIM and BF_4 – BF_4 appeared around 0.8 nm, indicating the alternating arrangement of ions with opposite charges. When the CNT was gradually charged, the nearest peaks of the RDFs based on ions with like charges (BMIM–BMIM and BF_4 – BF_4) tended to overlap with the nearest peak of the opposite ion pairs (BMIM– BF_4), indicating the appearance of ions with homogeneous charge in the nearest coordination shell around the central ions. Such results above reflect that the interactions between the ions changed in the CNTs with different charge densities, which could also be found from the interactions experienced by the ions confined in different CNTs (Table S1, ESI†). The RDFs between BMIM– BF_4 in different regions of the CNTs are shown in Fig. S4 (ESI†),

in which the nearest peaks could be seen to emerge at similar positions, while the peak intensities in the boundary region of the charged CNTs were stronger due to high local densities of the ILs.

We selected the representative hydrogen atoms (H_5) from the cations as an example, and calculated the RDFs between H_5 and the fluorine (F) from the anions, as shown in Fig. 5(d). The nearest peaks of the RDFs between H_5 and F emerged in similar positions; however, the peak intensity decreased with the increase in absolute charge density on the CNTs. The RDFs between the other hydrogen atoms from the cations and F atoms from the anions are presented in Fig. S5 (ESI†). The weak effect of the CNT charge on the spatial relationship of opposite ions was also reflected by the contact between the H atoms from each group in the cations and the F atoms from the anions. Nevertheless, the H-bond (HB) networks formed by the F atoms with strong electronegativity from the anions and H atoms strongly depended on the charge carried by the CNT (Fig. 5(e)). The way of defining the formation of HBs between ILs differs dramatically from that of water molecules, which typically need to satisfy both a bond length and a certain bond angle simultaneously, while the former one is more complex due to the formation of HBs by multiple H atoms.⁴¹ Herein, the bond length to define HBs was determined by the RDFs between the H atoms from each group in the cations and the F atoms (Fig. 5(d) and Fig. S5, ESI†), whereby a HB was

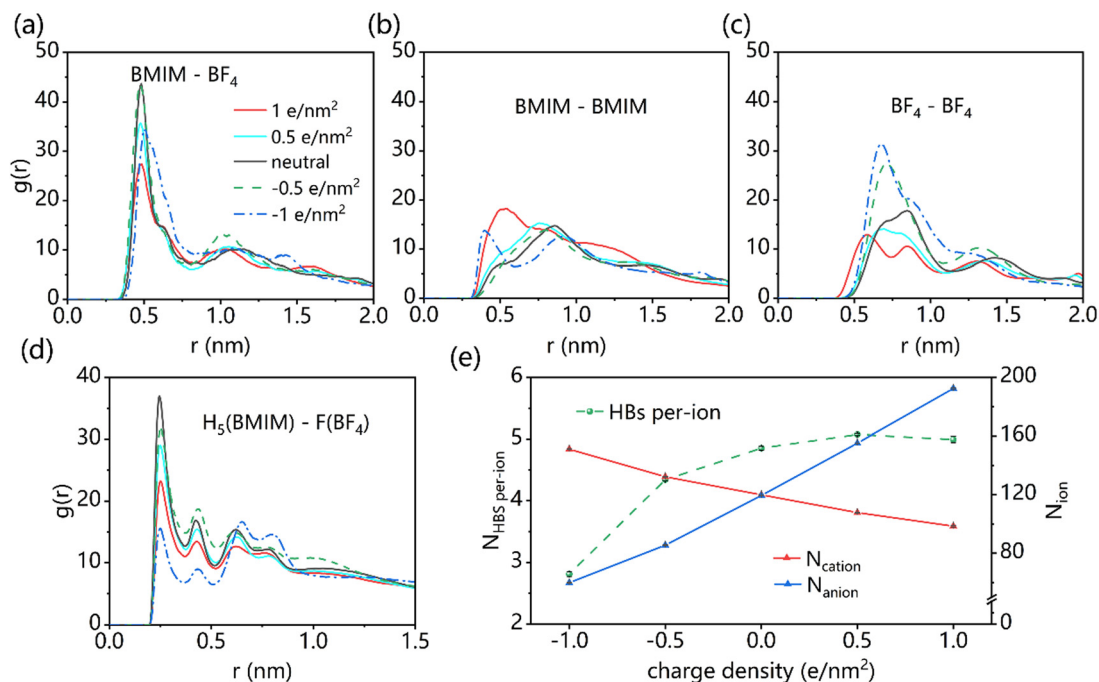


Fig. 5 Radial distribution functions (RDFs) between the COMs of (a) BMIM–BF₄, (b) BMIM–BMIM and (c) BF₄–BF₄ as a function of the distance r in CNTs with different charge densities. (d) RDFs of the H₅(BMIM)–F(BF₄) atoms pair (defined in Fig. 1(b)). (e) Numbers of cations (N_{cation}) and anions (N_{anion}) in the CNTs as a function of the charge density of the CNTs, based on the right axis. The average HBs number per ion formed by H atoms in cations and F atoms in anions in the CNTs, based on the left axis. The error bars were based on at least five independent simulations, and the sizes of error bars were smaller than symbols of data points.

considered to form if the distance between the H and F atoms was smaller than the first peak position of the RDF, while the bond angle effect was not included in the calculations for the HBs. As shown in Fig. 5(e), more positive charge imposed on the CNT led to slightly more HBs. On the contrary, a higher negative charge density on the CNT resulted in a more significant decline in HBs. First, the number of ions in the CNT decreased as the surface charge of the CNT switched from positive to negative, due to the asymmetric sized of the cations and anions. Then, according to the calculated numbers of ions in the CNT in Fig. 5(e) and the structural properties concluded above (in Fig. 2–4), the increase in HBs caused by the positive charge on the CNT depends on the rearrangement of the ion layers. In this situation, the cations were clamped by two anions layers from the surface direction and from the direction of the center of the CNT (Fig. 2(a) and (b)). In addition, both the butyl tails and imidazole rings of the cations were oriented more perpendicular to the tube wall with the growth of positive charge (Fig. 3(c)), which allowed more anions to contact with the butyl tails and imidazole rings, facilitating the formation of more HBs. On the other hand, the anions were clamped by neighboring cations layers when the imposed charge on the CNT was negative, although the vertical imidazole rings were still vertical to the tube wall, but the butyl tails were oriented parallel to tube wall, and the cations in Region C2 contacted with both the tube wall and the anions rather than touching the anions from both sides (Fig. 2(e)); while the vertical oriented imidazole rings in Region C1 located at the center of the CNT

(Fig. 4) also hindered the formation of HBs with anions. We also calculated the RDFs between different H atoms in cations and C atoms in CNTs with different charge densities, and the results are available in Fig. S6 (ESI†). Generally, the H atoms contacted with C atoms with larger probability if the imposed charge on the CNT was negative, which also resulted in a potential decrease in HBs between H atoms and F atoms. The different first peak positions of RDFs of H₅–C and H₄–C also indicated the perpendicular orientation of the imidazole rings in the cations in the CNT with -1 e nm^{-2} (Fig. S6, ESI†).

3.2 Region-dependent dynamics of the confined ILs

The above discussion demonstrated that the charge state of the CNT indeed influences the structure of ILs, so it could be inferred that such an influence may extend to the dynamics of confined ILs, and the heterogeneity of the structure in different regions of the CNT suggests the region-dependent dynamic properties of ILs. To measure the dynamics of ILs confined in a CNT, the diffusion coefficient was calculated *via* the following equation:

$$D = \frac{1}{2n} \lim_{\tau \rightarrow \infty} \frac{dr^2(\tau)}{d(\tau)} \quad (2)$$

where $r^2(\tau)$ is the mean squared displacement (MSD) of the center of mass of each ion under confinement, and n is the dimensionality for calculating the diffusion coefficient. The diffusion coefficient of ions was obtained from the slope of the fitted linear curve of the MSD *via* Einstein relation.

The MSD of N ions in the CNT is given by:

$$r^2(\tau) = \frac{1}{N} \sum_{i=1}^N \langle |r_i(t+\tau) - r_i(t)|^2 \rangle \quad (3)$$

where $r_i(t)$ is the position of ion i at time t , and $r_i(t+\tau)$ is the position of this ion at time $t+\tau$, where the absolute value of the difference between the former and the latter is the displacement of the ion within a certain period of time. Since the diffusion along the non-axial directions is strongly restricted in a CNT with a one-dimensional structure, only the diffusion along the axial direction z was calculated in this work, with the value of n in eqn (2) being 1. The calculated diffusion coefficients of ILs confined in CNTs with different charge densities were plotted in Fig. 6(a). When the charge densities of the CNTs were $\pm 0.5 \text{ e nm}^{-2}$, the diffusions of both cations and anions were faster than in the electroneutral CNT, while the diffusions were diminished when the charge densities were further increased to $\pm 1 \text{ e nm}^{-2}$. Such a non-monotonic behavior of the dynamics of ILs emerges owing to several factors jointly. The previous works presented that the potential impacts related to the dynamics included the number of ions in the nanotube, the intermolecular interactions, and the anchoring effect of ILs due to their interactions with the nanotube and so on, among which the intermolecular influences of ions were generalized as the interlocking of ions.^{22,42}

The numbers of cations and anions were close in the CNT with electroneutrality (Fig. 2(c)), so most ions were interlocked with those with opposite charge, leading to a quasi-crystal structure and slower diffusive dynamics of the ions.²² But when the charge densities of the CNTs went up to $\pm 0.5 \text{ e nm}^{-2}$, more counter-ions but less co-ions entered the CNTs and caused a non-equality between the cations and anions, thus breaking some interlocks between the cations and anions and weakening that hindrance to diffusion. To explain the interactions included in the interlocking of ions more directly, the values of the intermolecular interactions between ions were computed, including the interactions of BMIM–BF₄, BMIM–BMIM, and BF₄–BF₄; besides, the interactions between the CNT and ions were also considered, as shown in Table S1 (ESI†).

The growth of the anchoring effect induced by charging the CNT and the reduction of the interlocking due to the variation in Coulomb interactions play a trade-off role in diffusion; the moderate charge densities ($\pm 0.5 \text{ e nm}^{-2}$) correspond to lower limitations from both simultaneously. Furthermore, the interlocking of ions also involved the HB networks, and the orientation of cations may affect the dynamics of the ions indirectly as well. The number of HBs in positively charged CNT was more than in negatively charged CNT (Fig. 5(e)); thus, the dynamics of the ions would be slower in the positively charged CNT than in the negatively charged one in the case of identical absolute charge density (Fig. 6(a)). In addition, the orientations of the imidazole rings of the cations in CNTs with $\pm 0.5 \text{ e nm}^{-2}$ showed more significant randomness than those in the neutral CNT (Fig. 3(c)), which suggests the faster dynamics in such charge densities, concurring with the finding of a previous study.²³ The slower dynamics of ions in CNTs with higher charge densities ($\pm 1 \text{ e nm}^{-2}$) was due to the stronger Coulomb interaction between the highly charged CNTs and the counter-ions.

The transport of ions from one region to another (cross-region dynamics) was quantified by the probability of ions staying in the initial regions for a certain period of time, namely the survival probability, as given by:⁴³

$$P(t) = \frac{N_{j,h}(t+\tau)}{N_{j,h}(t)} \quad (4)$$

where $N_{j,h}(t)$ is the number of ions in region h at the initial time t , $N_{j,h}(t+\tau)$ is the number of ions remaining in this region after a period of time τ , and j is the ion species. As shown in Fig. 6(b) and (c), the survival probability of cations in the boundary region of the $+0.5 \text{ e nm}^{-2}$ CNT decayed most rapidly, indicating that the cations moved faster between the boundary and central regions in such a CNT than those with other charge densities; on the contrary, the cross-region movement was almost negligible in the CNT with -1 e nm^{-2} due to the presence of stable layers formed by the cations (Fig. 2(e)). For anions, the cross-region movement was also slowest in the CNT with the highest counter charge density $+1 \text{ e nm}^{-2}$; therefore, showing a similar behavior to the cations; whereas the fastest cross-region movement emerged in the neutral CNT. Such a difference indicates

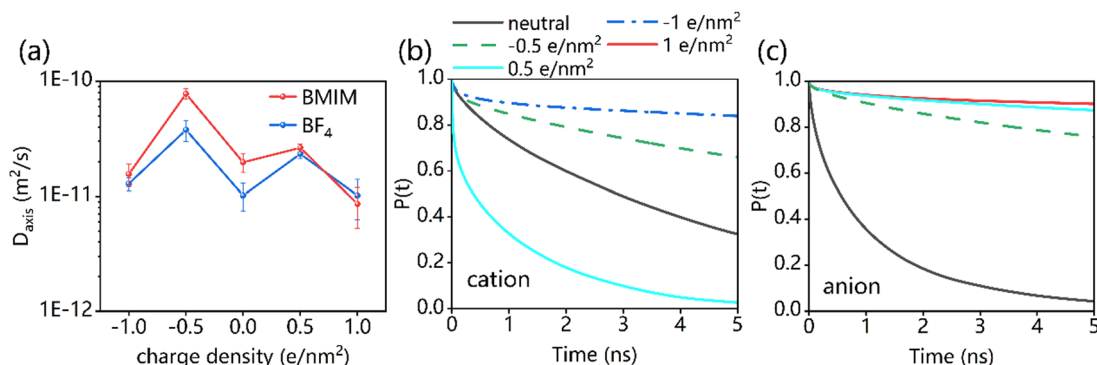


Fig. 6 (a) Diffusion coefficients along the axial direction as a function of the charge density of CNTs, where the error bars were based on at least five independent simulations; (b) and (c) survival probabilities of cations and anions in the boundary regions, where the co-ions of CNTs with $\pm 1 \text{ e nm}^{-2}$ were not considered because of single layers being formed in the CNTs.

that the cross-region movements of ions were not only related to the interactions with the CNT. The ions confined in the CNT can rarely transport across regions, owing to the strong anchoring effect of the ions on CNTs with a high counter charge density ($\pm 1 \text{ e nm}^{-2}$). However, the contribution of the ions themselves is also notable. We calculated the HB profiles in Region A1 and Region A2 in CNTs with different charge densities (Fig. 7(a)–(e)). Based on the distribution of HBs in the CNT with -0.5 e nm^{-2} (Fig. 7(b)), it was evident that the number of HBs around the interface between Region A1 and Region A2 was reduced to nearly 0, which indicates that if anions move from Region A2 to Region A1, they need to thoroughly break free from the original HB network in the region where they used to be located and subsequently construct a new HB network with ions in the other region, whereby the HB barrier hinders the cross-region movement of anions. In contrast, such a barrier was weakened owing to the gentler transition of the HB network across regions in the neutral CNT (Fig. 7(c)), allowing the faster transport of anions from one region to the other. The density distributions of the anions (Fig. S2(a)–(e), ESI†) show that the layer arrangement of anions was more significant than for cations, and the distinct distribution of anions in the CNT with -0.5 e nm^{-2} could also explain the slow cross-region dynamics (Fig. S2(d), ESI†). Likewise, the cross-region movement of cations could also be explained by the distribution of HBs (Fig. S7, ESI†).

The region-dependent dynamics of ions is based on the MSDs of ions located in different regions, which can be calculated according to the following equation:

$$r_{j,h,z}^2(\tau) = \frac{1}{N_{j,h}(t+\tau)} \sum_{i=1}^{N_{j,h}(t+\tau)} \langle |r_{j,h,z}(t) - r_{j,h,z}(t+\tau)|^2 \rangle \quad (5)$$

where the first subscript j of r denotes the species of ions, the second h denotes the region, and the third z represents the MSD calculated along the axial direction. The calculated region-dependent MSDs are plotted in Fig. 8(a)–(e). The heterogeneity of the ion dynamics in different regions of one-dimensional and cylindrical CNTs differ from that presented in two-dimensional slit pores, where the ion dynamics in the central region are typically faster than in the boundary region, regardless of the charge status of the pores.^{44,45} In this work, when the ILs were confined in cylindrical CNTs, the cations exhibited faster diffusion in the central region only when confined in CNTs with charge densities of $+0.5 \text{ e nm}^{-2}$ or neutral (Fig. 8(b) and (c)), while the anions diffused faster in the central region only in the neutral CNT (inset of Fig. 8(c)). The ions in the rest of the CNTs exhibited faster diffusions in the boundary regions. Such a difference related to the shape and dimensionality of nanochannels was similar to that presented by the region-dependent dynamics of water under confinement in nanochannels with different shapes and dimensionality.^{46,47} As discussed earlier, the ions in different regions showed discrepancies in both their local environments and constructions when confined in the CNTs, due to the different curvatures of the central and boundary regions in the CNTs; but the particles in different regions of the slit pores experienced analogous local environments owing to the flat geometry, resulting in similar structures of the two-dimensional slices, which may lead to the difference in the CNTs and slit pores. In addition, the RDFs of ions with like charges (BMIM–BMIM, BF_4 – BF_4) in different regions (Fig. S8, ESI†) indicated that the central ions were distributed more densely than in the boundary region in the neutral CNT

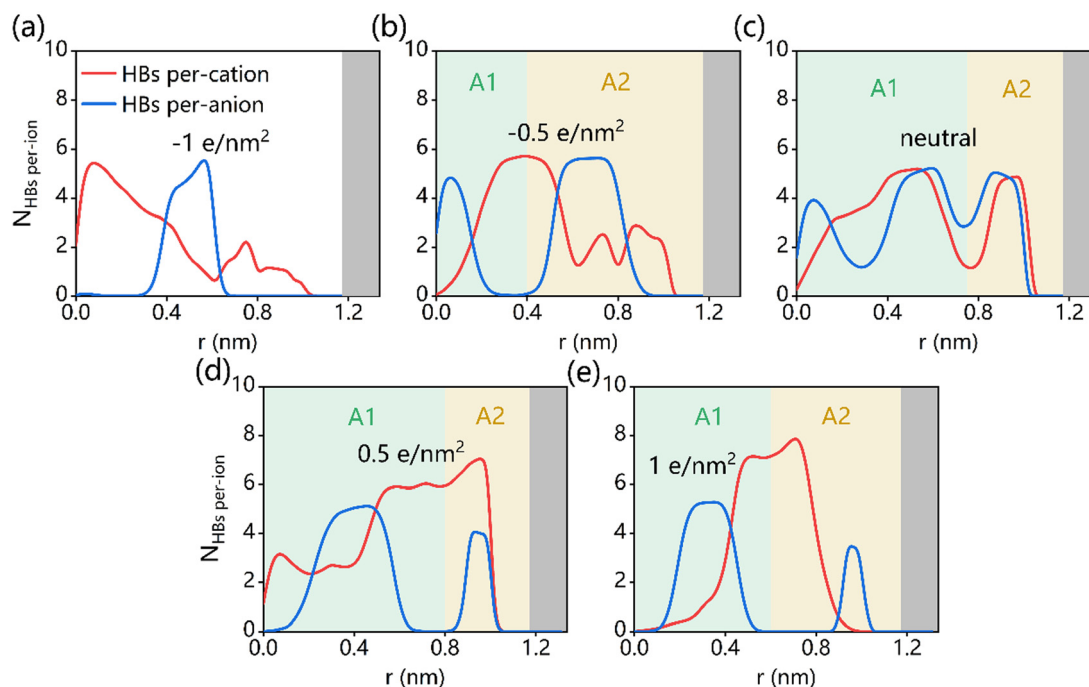


Fig. 7 Average HB number per cation (red lines) and per anion (blue lines) distributions along the radial direction in CNTs with charge densities of (a) -1 , (b) -0.5 , (c) 0 (neutral), (d) 0.5 , and (e) 1 e nm^{-2} , where the regions were divided based on the distribution of anions.

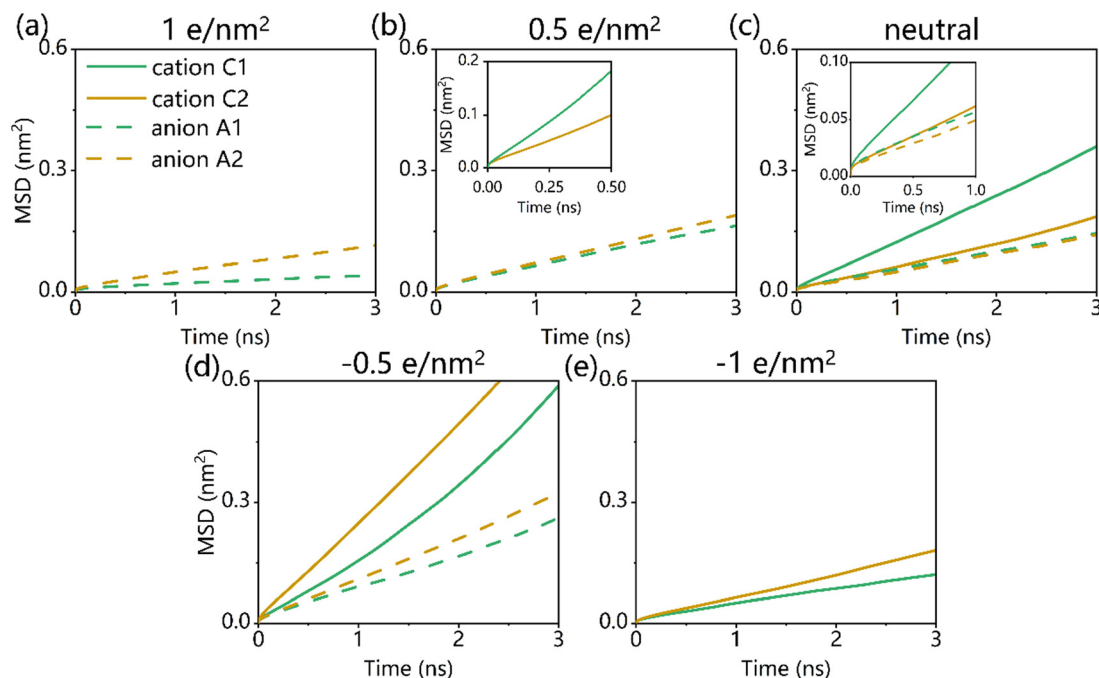


Fig. 8 Mean squared displacements (MSDs) along the axial direction of cations (solid lines) and anions (dashed lines) in the boundary regions and central regions of CNTs with charge densities of (a) 1, (b) 0.5, (c) 0 (neutral), (d) -0.5 , and (e) -1 e nm⁻². The inset in (b) shows the MSDs of cations calculated just for 0.5 ns to make sure the MSD was linear and credible; as time progresses, the MSD loses linearity and reliability due to the rare cations remaining in the original regions after a longer period.

(the positions of the first peaks in RDFs); while denser arrangements emerged in the boundary regions of the charged CNTs, reflecting the change in intermolecular interactions from the side direction.

To compare the interlocking of ions in different regions specifically and directly, the intermolecular interactions were computed and are compared in Fig. 9, with the full detailed results shown in Table S2 (ESI†). Since the difference in the electrostatic interactions of the ions in different regions of the same CNT with the tube wall was trivial compared to that in the differently charged CNTs (Table S2, ESI†), the heterogeneity of ion dynamics was mainly due to the interlocking effect of ions. We also measured the intermolecular interactions between the ions in bulk circumstance, as obtained from the bulk simulation with 600 IL pairs at an identical temperature. We found that the ions in a specific region diffused more rapidly when there was a larger difference in the total inter-ion interactions (the summations of the LJ and Coulomb interactions) between in the region and in the bulk. For instance, cations in Region C1 (central region) exhibited faster dynamics (inset in Fig. 8(b)) due to the greater gap between the total inter-ion interactions in the bulk and in the CNT with $+0.5$ e nm⁻²; on the other hand, the cations diffused more slowly in Region C2 (boundary region) of this CNT, attributed to the smaller gap. The difference in the value of the total inter-ion interactions presented by the ions in different regions compared to the bulk system reflects the breaking extent of the ion interlocks, if the anchoring effect on the CNT is similar. In other words, the ions have larger probabilities to diffuse freely if the ion interlocks are disrupted more significantly.

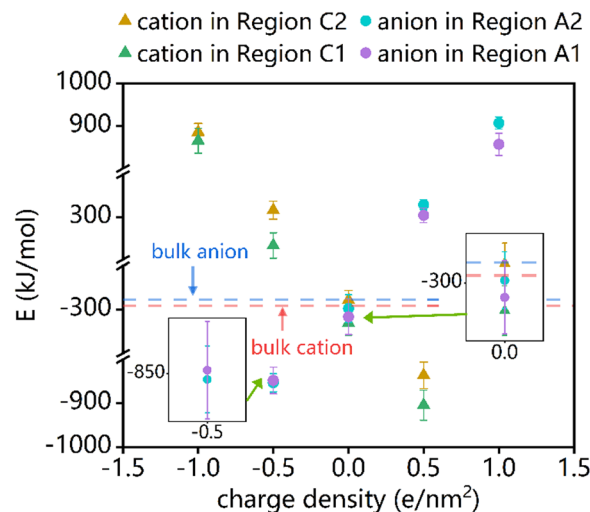


Fig. 9 Total intermolecular interactions between ions in the boundary regions and central regions of CNTs with different charge densities. The red and blue dashed lines represent the total intermolecular interactions of the bulk cations and anions, respectively. The data comparison of anions in Region A2 and Region A1 of -0.5 e nm⁻² CNT are magnified in the left inset, while the right one highlights a comparison of the data of both cations and anions in different regions of the neutral CNT.

3.3 Density of the vibrational states of ions

The impact of the charge state of the CNT on the structure of the confined IL was also characterized by the density of vibrational states (VDOS) of the ions in the CNT. The VDOS is

regarded as the Fourier transform of the velocity autocorrelation function of ions,⁴⁸ wherein the velocity autocorrelation function was calculated as the following:

$$C(\tau) = \left\langle m_i^{\frac{1}{2}} \mathbf{v}_i(t) \cdot m_i^{\frac{1}{2}} \mathbf{v}_i(t + \tau) \right\rangle \quad (6)$$

where $\mathbf{v}(t)$, $\mathbf{v}(t + \tau)$ are the velocities of atom i at times of t and $t + \tau$, respectively, and m_i is the mass of atom i . We calculated the VDOS of cations and anions separately to analyze the influence of the CNT charge states on the vibration modes of the ILs. The low-frequency range ($0\text{--}400\text{ cm}^{-1}$) of the VDOS is mainly related to the intermolecular vibration of ions, while the high-frequency range ($400\text{--}4000\text{ cm}^{-1}$) is derived from the intramolecular vibration.⁴⁸ The peak intensities of the VDOS of the cations and anions are shown in Fig. 10, and the full results are provided in Fig. S9 (ESI†). The low-frequency VDOS results of the counter-ions (ions with opposite charge to the CNTs) were shifted to higher frequencies when the ions were confined by CNTs with higher charge densities ($\pm 1\text{ e nm}^{-2}$), indicating an enhancement of the intermolecular vibration; while the co-ions exhibited a slight growth in the band intensity of the VDOS. The low-frequency VDOS results of the ions confined in the $\pm 0.5\text{ e nm}^{-2}$ CNTs in Fig. 10 show that the ions in the CNTs with lower charge densities exhibited less change in their intermolecular vibration mode, compared to the neutral CNT. The intermolecular vibration reflected by the low-frequency range of the VDOS is related to the structure and intermolecular interactions between ions.⁴⁹ In CNTs with higher charge densities, the higher total potential energies lead to an increase in vibrational frequency (Fig. 9 and Table S2, ESI†). Meanwhile, when the ions were confined in the CNTs with high counter charge densities, the original peaks in the RDFs between ion pairs with like charges were split into several narrower peaks with sharper shapes and nearer first peaks (Fig. 5(b) and (c)), indicating that the ions were arranged more orderly and tightly with those with like charges. Furthermore, the imidazole rings

and butyl tails of the cations were oriented almost similarly. Thus, it could be inferred that a higher counter charge density of the CNT may lead to a more highly ordered structure of ions, like crystals, causing a change in vibrations. For the high-frequency part with the wavenumber above 400 cm^{-1} in the VDOS results of the cation, all the bands could be assigned to intramolecular vibrations, including both the bending and stretching of imidazole rings and butyl tails.^{50,51} Wherein the medium frequency vibration modes ($400\text{--}2600\text{ cm}^{-1}$) were not affected by the charge states of the CNTs significantly, with only a few bands showing slight changes in intensity. However, several bands with lower wavenumber (roughly $2700\text{--}2850\text{ cm}^{-1}$) in the high-frequency part from 2600 cm^{-1} showed a slight blue-shift when the cations were confined in the CNT with -1 e nm^{-2} (inset of Fig. 10(b)). These bands could be assigned to the stretching of the bonds in the butyl tails and methyl group,⁵¹ and the blue-shift of the bands indicated a change in these vibrations. Moreover, the band at the frequency between $2900\text{--}3000\text{ cm}^{-1}$, corresponding to the three C–H bonds in the imidazole ring,⁵¹ was even split into a weaker main band and a tiny shoulder band with higher frequency. This could be due to the significant difference in the interaction between the three hydrogen atoms (H_5 , $\text{H}_4(\text{C}_1)$, $\text{H}_4(\text{C}_4)$) bonded to the imidazole ring and the CNT with -1 e nm^{-2} , compared with the other charged CNTs (Fig. S6, ESI†), resulting in a change in the vibrations of the bonds. The anions exhibited much fewer bands in the high-frequency part than the cations due to their simple intramolecular structure, wherein the band near 1000 cm^{-1} was assigned to the stretching of the B–F bonds in the anions.⁵² The single band was split into two weaker bands for the anions in the CNT with 1 e nm^{-2} , which may be related to the change in the stretching vibrations of B–F bonds caused by the significant changes in the intermolecular interaction of anions compared to the situation within the neutral CNT,⁵² although this band did not originate from the intermolecular vibrations.

3.4 Structures and HB networks in CNTs with different pore sizes

In order to explore how the nature of the IL under confinement correlates with the pore sizes of CNTs, we adopted neutral CNTs with different pore sizes. The density distributions of cations and anions in the CNTs with different pore sizes are shown in Fig. 11(a). The cations and anions were distributed one-dimensionally and alternately near the central axis of the (7,7) CNT. In the CNTs with chiral indices of 9, 10, and 11, the ions were distributed as ring structure coaxial with the CNTs, exhibiting a one-layer arrangement. The transition from single-layer to double-layer structures occurred in the (12,12) CNT, where a few cations were disengaged from the rings formed by the rest of the cations on the CNT surface and moved to the more central region, leading to the appearance of a weak peak near the central axis. When the pore size was further enlarged, more ions could enter the central region of the CNT, forming double layers distributed separately in the boundary regions and the central regions. The normalized orientation PDFs of the cationic imidazole rings in the different CNTs are compared in Fig. S10 (ESI†). When the ions were arranged as rings

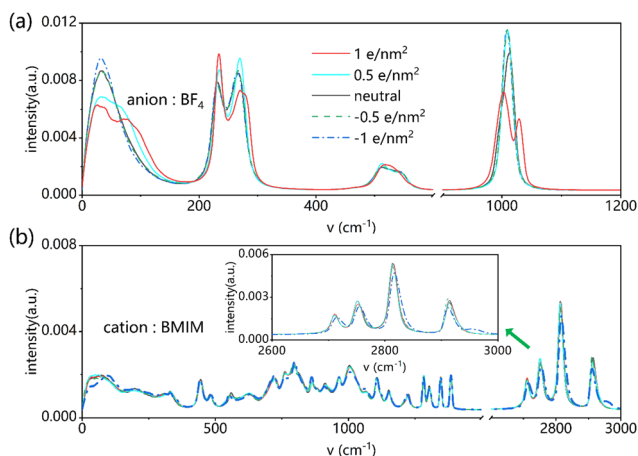


Fig. 10 Density of the vibrational states of the (a) anions and (b) cations in CNTs with different charge densities, with only the parts containing obvious bands shown in this figure. The complete results of the VDOS are plotted in Fig. S9 (ESI†).

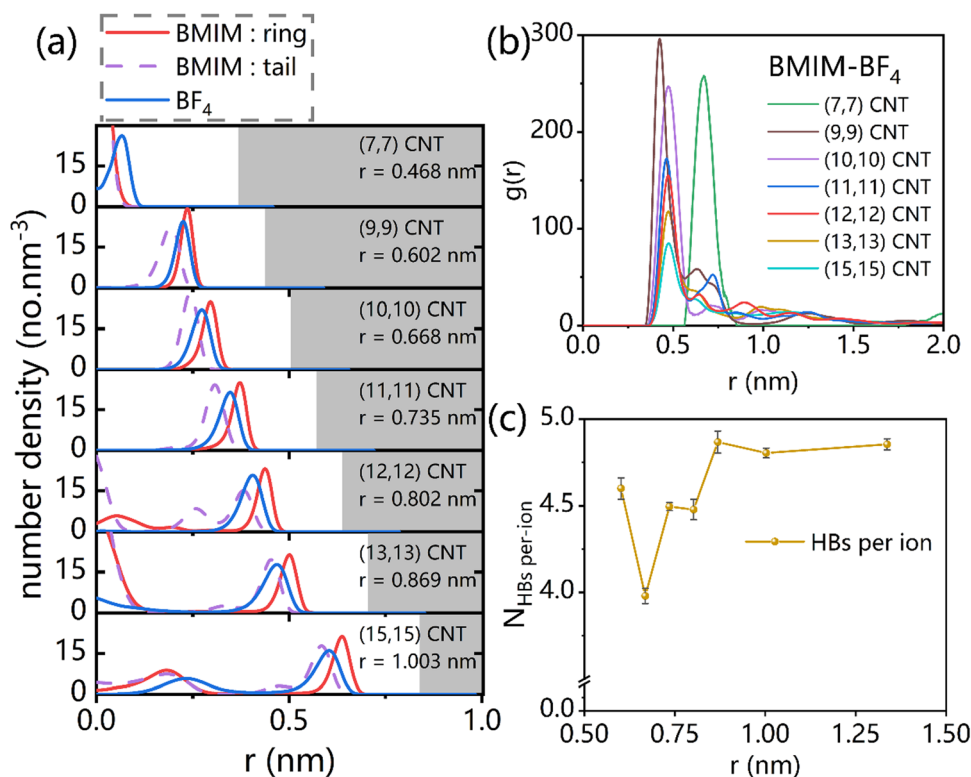


Fig. 11 (a) Number density profiles in CNTs with different pore sizes, where r represents the radius of each CNT. (b) RDFs between cation–anion pairs in adopted CNTs, as a function of the distance r . (c) Average number of HBs per ion as a function of the radii of the CNTs.

adjacent to the tube wall, then all the imidazole rings were oriented similarly due to the ion–wall correlation, with predominant orientations parallel to the tube wall; but the orientation PDF in the (7,7) CNT was very different, whereby the one-dimensionally arranged ions led to no obvious favorable orientation of the imidazole rings. The RDFs between the cations and anions in the CNTs with different pore sizes are exhibited in Fig. 11(b). In the narrowest (7,7) CNT, the first peak of RDF was shifted to the right-hand side compared to the wider CNTs, which could be attributed to the one-dimensional peculiar alternating arrangement of the ions due to the extremely strong confinement. The RDF in the (9,9) CNT showed the first maximum in a slightly left place, indicating the denser arrangement of ions. But when the pore size was further increased, the positions of the first peaks of RDFs did not change obviously, and just showed an abatement of the peak intensities. The layer structure of ions and their spatial relationship in CNTs with various pore sizes result in differences in the HB networks, as illustrated by the number of HBs in the different CNTs (Fig. 11(c)). (The ions in the (7,7) CNT were not considered to form HBs owing to the peculiar structure in this CNT). The non-monotonic behavior of the number of HBs with the pore size indicated that changes in the pore size result in complex changes to the microscopic ionic structure and spatial relationships. The numbers of HBs formed by different H atoms in the cations are exhibited in Fig. S11 (ESI[†]), which indicates that the fewer HBs formed in (10,10) CNT were due to the small amount of HBs formed by H atoms in the butyl tails.

The diffusions of ions in the narrower CNTs were not counted as too few ions existed in the CNTs for a period of time.

4. Conclusions

In conclusion, we explored the effect of the charge states of CNTs on the properties of ILs inside CNTs *via* MD simulations and discussed the influence of the pore size on the structures of ILs briefly. The anions and cations exhibited heterogeneous distributions within the CNTs owing to the charge densities of the CNTs. The orientation of the imidazole rings of the cations changed from parallel to vertical to the tube wall as the charge density of the CNT increased, yet the orientation of the butyl tails of the cations did not strongly depend on the charge state of the CNT. The state of ions changed between the alternating arrangement of opposite ions and the electric double-layer structures by altering the charge state of the CNT, as noted from the observations of the RDFs and density distributions, and the HBs were reduced by switching the surface charge of the CNT from positive to negative. The complicated dynamical behavior of confined ILs was observed to be due to the structural properties. The non-monotonic relationship between the charge densities of the CNTs and the diffusive coefficients of the ions suggested a trade-off effect on ionic diffusion from the abatement of the interlocking of the ions and the growth of the anchoring effect on the CNTs as the absolute charge densities increased. Besides, the differences in the local

environment and structure of the ions in the boundary regions and central regions led to a change in the breaking extents of the ion interlocks, resulting in the region-dependent dynamics of the ILs in cylindrical CNTs. The VDOS of the ions in CNTs with different charge densities also showed the impact of the charge on both the structures and interactions, corresponding to the results for the orientation and RDF. The ILs showed different properties in neutral CNTs with different pore sizes. As the pore sizes of the CNTs increased, the ILs exhibited different structures with varying numbers of layers; besides, the changes in the structure led to a non-monotonic tendency of the HB networks in CNTs with different pore sizes. The consequences of this work reveal the confinement effect of charged CNTs on the properties of ILs, providing insights into the relation between the transition of both the structure and dynamics and charge densities of CNTs, quantifying the interlocking of ions *via* intermolecular interactions and explaining the relation between the interlocking and ion dynamics. This work is meaningful for aiding finding the appropriate working states of electric devices with confined ILs and for optimizing their performance.

Conflicts of interest

There are no conflicts of interest to declare.

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References

- 1 R. Hayes, G. G. Warr and R. Atkin, *Chem. Rev.*, 2015, **115**, 6357–6426.
- 2 S. G. Zhang, J. H. Zhang, Y. Zhang and Y. Q. Deng, *Chem. Rev.*, 2017, **117**, 6755–6833.
- 3 S. Sahoo, K. Krishnamoorthy, P. Pazhamalai, V. K. Mariappan, S. Manoharan and S.-J. Kim, *J. Mater. Chem. A*, 2019, **7**, 21693–21703.
- 4 Y. Wang, G. Yang, F. Jiang, T. Qiu, Q. Liu, L. Zhou, C. Yang, J. Huang and G. Dai, *J. Mater. Chem. A*, 2022, **10**, 16795–16802.
- 5 I. Sullivan, A. Goryachev, I. A. Digdaya, X. Li, H. A. Atwater, D. A. Vermaas and C. Xiang, *Nat. Catal.*, 2021, **4**, 952–958.
- 6 T. Li, Y. Wang, S. Li, X. Liu and J. Sun, *Adv. Mater.*, 2020, **32**, 2002706.
- 7 V. K. Bhosale, J. Jeong and S. Kwon, *Fuel*, 2019, **255**, 115729.
- 8 J. H. Jun, J. Paeng, J. Kim, J. Shin, I.-S. Choi and J.-H. Lee, *ACS Appl. Mater. Interfaces*, 2023, **15**, 25484–25494.
- 9 R. R. Salunkhe, J. Lin, V. Malgras, S. X. Dou, J. H. Kim and Y. Yamauchi, *Nano Energy*, 2015, **11**, 211–218.
- 10 L. Yang, K. Zhuo, X. Xu, Z. Zhang, Q. Du, Y. Chen, D. Sun and J. Wang, *J. Mater. Chem. A*, 2020, **8**, 21789–21796.
- 11 Y.-H. Zou, Y.-B. Huang, D.-H. Si, Q. Yin, Q.-J. Wu, Z. Weng and R. Cao, *Angew. Chem., Int. Ed.*, 2021, **60**, 20915–20920.
- 12 A. Sagara, H. Yabe, X. Chen, B. Put, T. Hantschel, M. Mees, H. Arase, Y. Kaneko, A. Uedono and P. M. Vereecken, *ACS Appl. Mater. Interfaces*, 2021, **13**, 40543–40551.
- 13 J. Wu, *Chem. Rev.*, 2022, **122**, 10821–10859.
- 14 Q. Berrod, F. Ferdeghini, P. Judeinstein, N. Genevaz, R. Ramos, A. Fournier, J. Dijon, J. Ollivier, S. Rols, D. Yu, R. A. Mole and J. M. Zanotti, *Nanoscale*, 2016, **8**, 7845–7848.
- 15 M. Karagianni, L. Gkoura, A. Srivastava, A. Chatzichristos, N. Tsolakis, G. Romanos, S. Orfanidis, N. Panopoulos, S. Alhassan, D. Homouz, J. Hassan, M. Fardis and G. Papavassiliou, *Commun. Mater.*, 2023, **4**, 9.
- 16 C. Prehal, C. Koczwar, H. Amenitsch, V. Presser and O. Paris, *Nat. Commun.*, 2018, **9**, 4145.
- 17 R. Futamura, T. Iiyama, Y. Takasaki, Y. Gogotsi, M. J. Biggs, M. Salanne, J. Ségalini, P. Simon and K. Kaneko, *Nat. Mater.*, 2017, **16**, 1225–1232.
- 18 J. Ye, Y.-C. Wu, K. Xu, K. Ni, N. Shu, P.-L. Taberna, Y. Zhu and P. Simon, *J. Am. Chem. Soc.*, 2019, **141**, 16559–16563.
- 19 W.-Y. Tsai, P.-L. Taberna and P. Simon, *J. Am. Chem. Soc.*, 2014, **136**, 8722–8728.
- 20 S. Zhou, K. S. Panse, M. H. Motevaselian, N. R. Aluru and Y. Zhang, *ACS Nano*, 2020, **14**, 17515–17523.
- 21 C. L. Wang, Y. L. Wang, Y. M. Lu, H. Y. He, F. Huo, K. Dong, N. Wei and S. J. Zhang, *Phys. Chem. Chem. Phys.*, 2019, **21**, 12767–12776.
- 22 S. Kondrat, P. Wu, R. Qiao and A. A. Kornyshev, *Nat. Mater.*, 2014, **13**, 387–393.
- 23 T. Mo, S. Bi, Y. Zhang, V. Presser, X. Wang, Y. Gogotsi and G. Feng, *ACS Nano*, 2020, **14**, 2395–2403.
- 24 Z. Liu, S. Huang and W. Wang, *J. Phys. Chem. B*, 2004, **108**, 12978–12989.
- 25 Y.-L. Wang, F. U. Shah, S. Glavatskih, O. N. Antzutkin and A. Laaksonen, *J. Phys. Chem. B*, 2014, **118**, 8711–8723.
- 26 Y.-L. Wang, *J. Phys. Chem. B*, 2020, **124**, 7452–7466.
- 27 Y.-L. Wang, B. Li and A. Laaksonen, *Phys. Chem. Chem. Phys.*, 2021, **23**, 19435–19456.
- 28 U. Essmann, L. Perera, M. L. Berkowitz, T. Darden, H. Lee and L. G. Pedersen, *J. Chem. Phys.*, 1995, **103**, 8577–8593.
- 29 Y. Shim and H. J. Kim, *ACS Nano*, 2010, **4**, 2345–2355.
- 30 D. Bedrov, J.-P. Piquemal, O. Borodin, A. D. MacKerell, Jr., B. Roux and C. Schröder, *Chem. Rev.*, 2019, **119**, 7940–7995.
- 31 S. Kondrat, G. Feng, F. Bresme, M. Urbakh and A. A. Kornyshev, *Chem. Rev.*, 2023, **123**, 6668–6715.
- 32 L. Zeng, T. Wu, T. Ye, T. Mo, R. Qiao and G. Feng, *Nat. Comput. Sci.*, 2021, **1**, 725–731.
- 33 Z. Wang, Y. Yang, D. L. Olmsted, M. Asta and B. B. Laird, *J. Chem. Phys.*, 2014, **141**, 184102.
- 34 S. Bi, H. Banda, M. Chen, L. Niu, M. Chen, T. Wu, J. Wang, R. Wang, J. Feng, T. Chen, M. Dincă, A. A. Kornyshev and G. Feng, *Nat. Mater.*, 2020, **19**, 552–558.
- 35 B. Hess, C. Kutzner, D. van der Spoel and E. Lindahl, *J. Chem. Theory Comput.*, 2008, **4**, 435–447.
- 36 L. Martínez, R. Andrade, E. G. Birgin and J. M. Martínez, *J. Comput. Chem.*, 2009, **30**, 2157–2164.

- 37 K. S. Panse, H. Wu, S. Zhou, F. Zhao, N. R. Aluru and Y. Zhang, *J. Phys. Chem. Lett.*, 2022, **13**, 9464–9472.
- 38 S. Di Lecce, A. A. Kornyshev, M. Urbakh and F. Bresme, *ACS Nano*, 2020, **14**, 13256–13267.
- 39 R. M. Lynden-Bell, A. I. Frolov and M. V. Fedorov, *Phys. Chem. Chem. Phys.*, 2012, **14**, 2693–2701.
- 40 M. V. Fedorov and R. M. Lynden-Bell, *Phys. Chem. Chem. Phys.*, 2012, **14**, 2552–2556.
- 41 P. A. Hunt, C. R. Ashworth and R. P. Matthews, *Chem. Soc. Rev.*, 2015, **44**, 1257–1288.
- 42 Y. He, R. Qiao, J. Vatamanu, O. Borodin, D. Bedrov, J. Huang and B. G. Sumpter, *J. Phys. Chem. Lett.*, 2016, **7**, 36–42.
- 43 L. F. Mercier Franco, M. Castier and I. G. Economou, *J. Chem. Theory Comput.*, 2016, **12**, 5247–5255.
- 44 J. Gäding, G. Tocci, M. Busch, P. Huber and R. H. Meißner, *J. Chem. Phys.*, 2022, **156**, 064703.
- 45 R. Singh, N. N. Rajput, X. He, J. Monk and F. R. Hung, *Phys. Chem. Chem. Phys.*, 2013, **15**, 16090–16103.
- 46 M. Zokaie and M. Foroutan, *RSC Adv.*, 2015, **5**, 39330–39341.
- 47 A. Barati Farimani and N. R. Aluru, *J. Phys. Chem. B*, 2011, **115**, 12145–12149.
- 48 V. H. Paschoal, L. F. O. Faria and M. C. C. Ribeiro, *Chem. Rev.*, 2017, **117**, 7053–7112.
- 49 S. S. Sarangi, S. K. Reddy and S. Balasubramanian, *J. Phys. Chem. B*, 2011, **115**, 1874–1880.
- 50 R. Holomb, A. Martinelli, I. Albinsson, J. C. Lassègues, P. Johansson and P. Jacobsson, *J. Raman Spectrosc.*, 2008, **39**, 793–805.
- 51 N. E. Heimer, R. E. Del Sesto, Z. Meng, J. S. Wilkes and W. R. Carper, *J. Mol. Liq.*, 2006, **124**, 84–95.
- 52 T. Kato, N. Nanbu, Y. Sasaki, T. Ohsaka and F. Kitamura, *Electrochemistry*, 2005, **73**, 589–592.